

Artificial Intelligence & Machine Learning – Task 4

Classification Models, Evaluation Metrics & Handling Imbalanced Data

1. Objective

The objective of this task was to develop a binary classification model and evaluate its performance using appropriate classification metrics. Unlike regression problems, where the output is a continuous value, classification problems predict discrete class labels. This task focused on understanding classification techniques, confusion matrix analysis, precision, recall, F1-score, ROC-AUC, and handling class imbalance.

2. Dataset Description

The Breast Cancer Wisconsin Dataset available in Scikit-learn was used for this project. The dataset contains diagnostic measurements of breast cancer tumors and is commonly used for binary classification problems.

Dataset Characteristics

- Total Samples: 569
- Total Features: 30
- Target Classes:
 - 0 = Malignant
 - 1 = Benign

The dataset contains features such as radius, texture, perimeter, area, smoothness, compactness, and other tumor characteristics.

3. Data Preprocessing

Several preprocessing steps were performed before model training:

3.1 Data Loading

The dataset was loaded using the `load_breast_cancer()` function from Scikit-learn.

3.2 Missing Value Analysis

The dataset was examined for missing values and no missing observations were found.

3.3 Train-Test Split

The dataset was divided into training and testing sets using an 80:20 ratio.

- Training Data: 455 samples
- Testing Data: 114 samples

Stratified sampling was applied to preserve the original class distribution in both training and testing datasets.

3.4 Feature Scaling

StandardScaler was used to standardize feature values.

Benefits of scaling:

- Improves model convergence
 - Prevents features with larger magnitudes from dominating the model
 - Enhances Logistic Regression performance
-

4. Logistic Regression Model

Logistic Regression was selected as the baseline classification model because it is widely used for binary classification tasks and provides interpretable results.

The model was trained using:

- Algorithm: Logistic Regression
- Maximum Iterations: 1000

After training, predictions were generated on the test dataset and evaluated using multiple classification metrics.

5. Evaluation Metrics

5.1 Accuracy

Accuracy measures the proportion of correctly classified observations.

Accuracy Formula:

$$\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN)$$

Result

Accuracy = 98.25%

Although accuracy was very high, additional metrics were evaluated because accuracy alone may be misleading in classification problems.

5.2 Confusion Matrix

The confusion matrix provides a detailed breakdown of model predictions.

It consists of:

- True Positives (TP)
- True Negatives (TN)
- False Positives (FP)

- False Negatives (FN)

The confusion matrix helps identify the types of errors made by the model and provides deeper insight than accuracy alone.

5.3 Precision

Precision measures how many positive predictions were actually correct.

Precision Formula:

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

Result

Precision values ranged from 0.98 to 0.99.

A high precision score indicates that the model generates very few false positive predictions.

5.4 Recall

Recall measures how many actual positive samples were correctly identified.

Recall Formula:

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

Result

Recall values ranged from 0.98 to 0.99.

Recall is particularly important in medical diagnosis because missing a cancer case can have severe consequences.

5.5 F1-Score

F1-score provides a balance between precision and recall.

Formula:

$$\text{F1} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

Result

F1-score values ranged from 0.98 to 0.99.

The high F1-score demonstrates strong overall classification performance.

6. ROC Curve and AUC Score

The Receiver Operating Characteristic (ROC) Curve was plotted to evaluate the classifier's discrimination capability across different classification thresholds.

AUC Score

AUC = 0.9954

Interpretation

An AUC value close to 1 indicates excellent classification performance.

The obtained AUC score of 0.9954 shows that the Logistic Regression model can effectively distinguish between benign and malignant tumors with very high reliability.

7. Handling Class Imbalance

Class imbalance is a common challenge in real-world machine learning applications.

To address this issue, Logistic Regression was also trained using:

```
class_weight = "balanced"
```

This technique automatically assigns higher weights to minority classes and lower weights to majority classes.

Benefits:

- Improves minority class detection
- Reduces bias toward majority classes
- Produces fairer classification performance

Although the Breast Cancer dataset was only slightly imbalanced, this approach demonstrates an industry-standard technique for handling imbalance.

8. Model Comparison

To compare performance, a Decision Tree Classifier was also trained and evaluated.

Logistic Regression Results

- Accuracy: 98.25%
- AUC: 0.9954
- F1-Score: 0.98–0.99

Decision Tree Results

- Accuracy: 91.23%
- F1-Score: 0.89–0.93

Comparison

Logistic Regression significantly outperformed the Decision Tree classifier.

Reasons:

- Better generalization capability
- Lower risk of overfitting
- Higher classification accuracy
- Superior ROC-AUC performance

Therefore, Logistic Regression was selected as the final model.

9. Cross-Validation Analysis

Five-fold cross-validation was performed to evaluate model stability.

Cross-Validation Scores

- 96.70%
- 97.80%
- 96.70%
- 100.00%
- 98.90%

Mean Cross-Validation Accuracy

98.02%

The cross-validation accuracy was very close to the test accuracy of 98.25%, indicating that the model generalizes well and does not suffer from significant overfitting.

10. Conclusion

In this task, a binary classification system was developed using the Breast Cancer Wisconsin Dataset. Logistic Regression and Decision Tree classifiers were trained and evaluated using multiple performance metrics.

The Logistic Regression model achieved:

- Accuracy: 98.25%
- AUC Score: 0.9954
- Cross-Validation Accuracy: 98.02%

Performance was assessed using confusion matrix, precision, recall, F1-score, ROC curve, and AUC score. The results demonstrate that Logistic Regression provides excellent predictive performance and strong generalization ability.

Based on the experimental results, Logistic Regression was selected as the final model because it achieved superior accuracy, higher stability, better discrimination capability, and lower overfitting risk compared to the Decision Tree classifier.

